

Cyrus Guided Notes: MIT 6.003 Signals and Systems

MIT course pack

2026-05-15

Course source

Official source: <https://ocw.mit.edu/courses/6-003-signals-and-systems-fall-2011/>

Why this course is in the system. Prerequisite for signals, systems, convolution, frequency response, filtering, and state space.

Learning output. Write the time-domain to frequency-domain bridge in GoodNotes and add convolution visualization on the web.

Prerequisite checkpoint

- Complex numbers
- Calculus
- Differential equations

Formula checkpoint

Do not memorize these first. Read each formula as a sentence: what changes, what is observed, what is optimized, and what output it produces.

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau)d\tau$$

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt$$

$$H(s) = \frac{Y(s)}{X(s)}$$

$$\dot{\mathbf{x}} = A\mathbf{x} + B\mathbf{u}$$

Visual page

Show an input signal sliding across an impulse response to create convolution.

GoodNotes pages

- Page MIT-S001: convolution
- Page MIT-S002: frequency response

Web interaction

A slider for frequency that updates magnitude and phase.

Obsidian and Notion

Layer	Output
Obsidian	MIT EECS -> 6.003 signals systems
Notion	Signal exercise and formula status.

First study move

Open the official source, copy the prerequisite checkpoint into GoodNotes, then write one formula in your own words before watching or reading more.